

Module 1: Introduction to Bootstrap

1. What is Bootstrap?

Bootstrap is a free and open-source CSS framework used to create responsive and mobile-friendly websites quickly.

It provides ready-made:

- Buttons
- Forms
- Navigation Bars
- Cards
- Modals
- Alerts
- Grid System
- Tables
- Responsive Layouts

Instead of writing CSS from scratch, Bootstrap provides pre-designed classes that make web development faster and easier.

Why Use Bootstrap?

Faster Development

Mobile Responsive Design

Easy to Learn

Cross-Browser Compatible

Professional UI Components

Large Community Support

Real Life Example

Without Bootstrap:

```
<button style="
background:blue;
color:white;
padding:10px 20px;
border:none;
border-radius:5px;">
Click Me
</button>
```

With Bootstrap:

```
<button class="btn btn-primary">
Click Me
</button>
```

Bootstrap automatically applies styling.

Features of Bootstrap

1. Responsive Design

Website automatically adjusts according to screen size.

Device	Screen Size
Mobile	Small
Tablet	Medium

Device	Screen Size
Laptop	Large
Desktop	Extra Large

2. Grid System

Bootstrap uses a 12-column layout system.

Example:

```
<div class="row">  
  <div class="col-6">Left</div>  
  <div class="col-6">Right</div>  
</div>
```

Output:

| Left | Right |

3. Ready-Made Components

Bootstrap provides:

- Buttons
- Forms
- Cards
- Navbar
- Modal
- Alerts
- Carousel

Example:

```
<div class="alert alert-success">  
Data Saved Successfully!  
</div>
```

Bootstrap Versions

Version	Release
Bootstrap 3	Old
Bootstrap 4	Popular
Bootstrap 5	Latest & Recommended

Current learning recommendation:

Bootstrap 5

Installation of Bootstrap

There are two main ways:

Method 1: Using CDN (Recommended for Beginners)

What is CDN?

CDN (Content Delivery Network) is an online server that provides Bootstrap files directly.

No download required.

Step 1: Create HTML File

```
<!DOCTYPE html>
<html>
<head>
<title>Bootstrap Demo</title>
</head>
<body>

<h1>Hello Bootstrap</h1>

</body>
</html>
```

Step 2: Add Bootstrap CDN

```
<!DOCTYPE html>
<html>
<head>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css
/bootstrap.min.css" rel="stylesheet">

<title>Bootstrap Demo</title>

</head>
<body>

<h1 class="text-primary">
Hello Bootstrap
</h1>
```

```
</body>  
</html>
```

Practical Example

```
<!DOCTYPE html>  
<html>  
<head>  
  
<link  
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css  
/bootstrap.min.css" rel="stylesheet">  
  
</head>  
<body>  
  
<div class="container mt-5">  
  
<h1 class="text-danger">  
Welcome to Bootstrap  
</h1>  
  
<button class="btn btn-success">  
Click Here  
</button>  
</div>  
</body>  
</html>
```

Output:

✓ Red Heading

✓ Green Button

✓ Professional Design

Adding Bootstrap JavaScript

Some components like:

- Modal
- Carousel
- Dropdown
- Tooltip

require JavaScript.

Add before closing body tag:

```
<script  
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/b  
ootstrap.bundle.min.js"></script>
```

Complete Setup:

```
<!DOCTYPE html>
```

```
<html>
```

```
<head>
```

```
<link
```

```
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css  
/bootstrap.min.css" rel="stylesheet">
```

```
</head>
```

```
<body>
```

```
<h1 class="text-primary">
```

```
Bootstrap Ready
```

```
</h1>
```

```
<script
```

```
src="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/js/b  
ootstrap.bundle.min.js"></script>
```

```
</body>
```

```
</html>
```

Method 2: Using npm

What is npm?

npm stands for:

Node Package Manager

Used in React, Angular, Vue, MERN projects.

Step 1: Install Node.js

Download and install Node.js.

Check installation:

```
node -v
```

```
npm -v
```

Step 2: Create Project

```
mkdir bootstrap-project  
cd bootstrap-project
```

Step 3: Initialize npm

```
npm init -y
```

Output:

```
{  
  "name": "bootstrap-project",  
  "version": "1.0.0"  
}
```

Step 4: Install Bootstrap

```
npm install bootstrap
```

Step 5: Import Bootstrap

For JavaScript Project:

```
import 'bootstrap/dist/css/bootstrap.min.css';
```

For Bootstrap JS:

```
import 'bootstrap/dist/js/bootstrap.bundle.min.js';
```

React Example

```
import 'bootstrap/dist/css/bootstrap.min.css';
```

```
function App() {  
  return (  
    <div className="container">  
      <h1 className="text-primary">  
        Hello Bootstrap  
      </h1>  
    </div>  
  );  
}
```

```
export default App;
```

CDN vs npm

Feature	CDN	npm
Easy Setup	?	?
Beginner Friendly	?	?
Internet Required	?	?
React Support	?	?
Angular Support	?	?
Vue Support	?	?
Professional Projects	?	?

Practical Activity 1

Create a Blue Button

```
<!DOCTYPE html>
<html>
<head>

<link
href="https://cdn.jsdelivr.net/npm/bootstrap@5.3.3/dist/css
/bootstrap.min.css" rel="stylesheet">
</head>
<body>
<button class="btn btn-primary">
Login
</button>
</body>
</html>
```

Practical Activity 2

Create a Responsive Container

```
<div class="container">
<h1>Welcome</h1>
<p>Bootstrap Learning</p>
</div>
```

Practical Activity 3

Create Alert Message

```
<div class="alert alert-success">
Registration Successful
</div>
```

Interview Questions

Q1. What is Bootstrap?

Bootstrap is a CSS framework used to create responsive and mobile-friendly websites quickly.

Q2. What are the advantages of Bootstrap?

- Responsive Design
 - Faster Development
 - Ready Components
 - Easy Customization
 - Cross-Browser Support
-

Q3. What is CDN?

CDN (Content Delivery Network) is an online server that provides Bootstrap files without downloading them.

Q4. What is npm?

npm (Node Package Manager) is used to install Bootstrap and other packages in modern web projects.

Q5. Which Bootstrap version should we learn?

Bootstrap 5 because it is the latest and most widely used version.

Module Summary

- ✓ Bootstrap is a CSS framework.
- ✓ It helps create responsive websites quickly.
- ✓ Bootstrap provides ready-made classes and components.
- ✓ Bootstrap can be installed using **CDN** or **npm**.
- ✓ CDN is best for beginners.
- ✓ npm is best for React, Angular, Vue, and MERN projects.
- ✓ Bootstrap 5 is the recommended version for learning and professional development.